Computabilidad y Complejidad

Ejemplos de máquinas de Turing

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Ejemplo 1

$$L(M) = \{a^n b^n : n \geq 1\}$$

$$M = (\{a, b\}, \{a, b, X, Y, B\}, \{q_0, q_1, q_2, q_3, q_4\}, f, B, q_0, \{q_4\})$$

	a	b	X	Y	B
q_0	(q_1, X, R)	———	———	(q_3, Y, R)	———
q_1	(q_1, a, R)	(q_2, Y, L)	———	(q_1, Y, R)	———
q_2	(q_2, a, L)	———	(q_0, X, R)	(q_2, Y, L)	———
q_3	———	———	———	(q_3, Y, R)	(q_4, B, R)
q_4	———	———	———	———	———

Ejemplo 1

	a	b	X	Y	B
q_0	(q_1, X, R)	—	—	(q_3, Y, R)	—
q_1	(q_1, a, R)	(q_2, Y, L)	—	(q_1, Y, R)	—
q_2	(q_2, a, L)	—	(q_0, X, R)	(q_2, Y, L)	—
q_3	—	—	—	(q_3, Y, R)	(q_4, B, R)
q_4	—	—	—	—	—

$$Bq_0aaabbb \vdash Xq_1aabbb \vdash Xaq_1abbb \vdash Xaaq_1bbb \vdash Xaq_2aYbb \vdash Xq_2aaYbb \vdash q_2XaaYbb$$

$$Xq_0aaYbb \vdash XXq_1aYbb \vdash XXaq_1Ybb \vdash XXaYq_1bb \vdash XXaq_2YYb \vdash XXq_2aYYb \vdash Xq_2XaYYb$$

$$XXq_0aYYb \vdash XXXq_1YYb \vdash XXXYq_1Yb \vdash XXXYYq_1b \vdash XXXYq_2YY \vdash XXXq_2YYY \vdash XXq_2XYYY$$

$$XXXq_0YYY \vdash XXXYq_3YY \vdash XXXYYq_3Y \vdash XXXYYYq_3B \vdash XXXYYYBq_4B$$

Ejemplo 2

$$L(M) = \{wcw^r : w \in (a+b)^*\}$$

$$M = (\{a,b,c\},\{a,b,c,B\},\{q_0,q_a,q_{1a},q_{2a},q_b,q_{1b},q_{2b},q_r,q_t,q_f\},f,B,q_0,\{q_f\})$$

	a	b	c	B
q_0	(q_a,B,R)	(q_b,B,R)	(q_t,B,R)	———
q_a	(q_a,a,R)	(q_a,b,R)	(q_{1a},c,R)	———
q_{1a}	(q_{1a},a,R)	(q_{1a},b,R)	———	(q_{2a},B,L)
q_{2a}	(q_r,B,L)	———	———	———
q_b	(q_b,a,R)	(q_b,b,R)	(q_{1b},c,R)	———
q_{1b}	(q_{1b},a,R)	(q_{1b},b,R)	———	(q_{2b},B,L)
q_{2b}	———	(q_r,B,L)	———	———
q_r	(q_r,a,L)	(q_r,b,L)	(q_r,c,L)	(q_0,B,R)
q_t	———	———	———	(q_f,B,R)
q_f	———	———	———	———

Ejemplo 3

$$\textbf{resta}(\boldsymbol{m},\boldsymbol{n}) = \begin{cases} m-n & \text{si} \quad m \geq n \\ 0 & \text{si} \quad m < n \end{cases}$$

$$\boldsymbol{M} = (\{\mathbf{0},\mathbf{1}\},\{\mathbf{0},\mathbf{1},\boldsymbol{B}\},\{\boldsymbol{q_0},\boldsymbol{q_1},\boldsymbol{q_2},\boldsymbol{q_3},\boldsymbol{q_4},\boldsymbol{q_5},\boldsymbol{q_6}\},\boldsymbol{f},\boldsymbol{B},\boldsymbol{q_0},\emptyset)$$

	0	**1**	**B**
q_0	(q_1, B, R)	(q_5, B, R)	———
q_1	$(q_1, 0, R)$	$(q_2, 1, R)$	———
q_2	$(q_3, 1, L)$	$(q_2, 1, R)$	(q_4, B, L)
q_3	$(q_3, 0, L)$	$(q_3, 1, L)$	(q_0, B, R)
q_4	$(q_4, 0, L)$	(q_4, B, L)	$(q_6, 0, R)$
q_5	(q_5, B, R)	(q_5, B, R)	(q_6, B, R)
q_6	———	———	———

Ejemplo 3

	0	1	B
q_0	(q_1, B, R)	(q_5, B, R)	—
q_1	$(q_1, 0, R)$	$(q_2, 1, R)$	—
q_2	$(q_3, 1, L)$	$(q_2, 1, R)$	(q_4, B, L)
q_3	$(q_3, 0, L)$	$(q_3, 1, L)$	(q_0, B, R)
q_4	$(q_4, 0, L)$	(q_4, B, L)	$(q_6, 0, R)$
q_5	(q_5, B, R)	(q_5, B, R)	(q_6, B, R)
q_6	—	—	—

$$Bq_0000100 \vdash Bq_100100 \vdash 0q_10100 \vdash 00q_1100 \vdash 001q_200 \vdash 00q_3110 \vdash 0q_30110 \vdash$$

$$Bq_300110 \vdash q_3B00110 \vdash Bq_000110 \vdash Bq_10110 \vdash 0q_1110 \vdash 01q_210 \vdash 011q_20 \vdash$$

$$01q_311 \vdash 0q_3111 \vdash Bq_30111 \vdash q_3B0111 \vdash Bq_00111 \vdash Bq_1111 \vdash 1q_211 \vdash$$

$$11q_21 \vdash 111q_2B \vdash 11q_41 \vdash 1q_41 \vdash Bq_41 \vdash Bq_4B \vdash 0q_6B$$

Ejemplo 4

$$multiplicacion(x, y) = x \cdot y$$

(con valores $x, y > 0$)

	0	1	#	B
q_{ini}	$(q_{ini2}, \#, R)$			
q_{ini2}	$(q_{ini2}, 0, R)$	$(q_{ini2}, 1, R)$		$(q_{ini3}, 1, L)$
q_{ini3}	$(q_{ini3}, 0, L)$	$(q_{ini3}, 1, R)$	(q_6, B, R)	
q_0	(q_6, B, R)			
q_1	$(q_2, \#, R)$	$(q_4, 1, L)$		
q_2	$(q_2, 0, R)$	$(q_2, 1, R)$		$(q_3, 0, L)$
q_3	$(q_3, 0, L)$	$(q_3, 1, L)$	$(q_1, \#, R)$	
q_4		$(q_5, 1, R)$	$(q_4, 0, L)$	
q_5	$(q_7, 0, L)$			
q_6	$(q_6, 0, R)$	$(q_1, 1, R)$		
q_7		$(q_8, 1, L)$		
q_8	$(q_9, 0, L)$			(q_{10}, B, R)
q_9	$(q_9, 0, L)$			(q_0, B, R)
q_{10}		(q_{11}, B, R)		
q_{11}	(q_{11}, B, R)	(q_{12}, B, R)		
q_{12}				